

Advanced Signal Processing

Scalar Kalman Filtering

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1 Scalar LMMSEE for First-order AR Process

In this problem we consider the estimation of a random process $\{Y(n)\}$ from observations of a random process $\{X(n)\}$. We model the (unobservable) random process $\{Y(n)\}$ as a first-order autoregressive (AR) process

$$Y(n) = h \cdot Y(n-1) + Z(n), \quad n = 1, 2, \dots, \quad (1)$$

where $\{Z(n)\} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \sigma_Z^2)$ and h a known deterministic constant. The random variable $Y(0) \sim \mathcal{N}(0, \sigma_Y^2)$ is assumed independent of $\{Z(n)\}$. To ensure that $\{Y(n)\}$ is WSS, we set $\sigma_Y^2 = \sigma_Z^2 / (1 - h^2)$. The observable process $\{X(n)\}$ is modeled as

$$X(n) = Y(n) + W(n), \quad n = 1, 2, \dots, \quad (2)$$

with $\{W(n)\} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \sigma_W^2)$.

1. Implement the model and plot realizations of $\{Y(n)\}$ and $\{X(n)\}$ for different settings of h , σ_Z^2 and σ_W^2 . Compare and discuss the results. What happens if h is negative?

For each n we now use a linear minimum mean square error estimator (LMMSEE) to estimate $Y(n)$ based on the observations of $X(1), \dots, X(n)$. From now on, we set $h = 0.95$, $\sigma_Z^2 = 0.1$ and $\sigma_W^2 = 1$.

2. Derive the LMMSEE for $Y(n)$ given the data $X(1), \dots, X(n)$. (*Hint: recall that the ACF of the AR process in (1) is given by $R_Y(k) = \sigma_Z^2 \frac{h^{|k|}}{1-h^2}$.*)
3. Implement the LMMSEE in a flexible way such that you can test it for different settings of n . Plot and compare the simulated mean square error (MSE) to the theoretical MSE for $n = 1, \dots, 30$. (*Hint: The `toeplitz` command in MATLAB may be useful to generate covariance matrices.*)
4. Push your simulator to the extreme — how many observations can you use before the simulator breaks down or gets too slow? What causes this effect?

2 Scalar Kalman Filter for First-order AR Process

We consider the same models as given in (1) and (2) in the previous problem. We set the parameters

$$h = 0.95, \quad \sigma_Z^2 = 0.1 \quad \sigma_W^2 = 1 \quad (3)$$

For each n we now use a Kalman filter to estimate $Y(n)$ based on the observation of $X(1), \dots, X(n)$.

1. Draw a block diagram of the state and observation model. State the equations for the prediction step and the updating step of the Kalman filter.
2. Next, write up (by hand) the prediction and updating steps, including the Kalman gain and the MSEs, for times $n = 1$ and $n = 2$ assuming that the first two observations received have values $X(1)$ and $X(2)$. Avoid inserting the numerical values of the parameters given in (3) and use their notations instead. Choose as well the initial values $\hat{Y}(0|0)$ and $R(0|0)$.
3. Implement the Kalman filter in Matlab and investigate the behavior of the simulated mean squared estimation error as a function of n by running multiple simulations. Compare to the theoretical MSE $R(n|n)$ and to the results for the LMMSEE in Problem 1. (*Hint: Refer to the lecture notes for initialization of the Kalman filter*).
4. Run the Kalman filter for large n values. Compare to the LMMSEE case.
5. Vary the parameters h , σ_Z^2 and σ_W^2 and investigate how it affects $R(n|n)$.
6. Initialize the filter differently i.e. with $\hat{Y}(0|0) = 100$ and $R(0|0) = 0.0001$. Investigate how it affects $R(n|n)$.
7. Calculate the steady state solutions for the MSE and compare to the simulated MSE for your Kalman filter.